

LG ABS XR401

Acrylonitrile Butadiene Styrene
LG Chem Ltd.

PROSPECTOR®

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Technical Data

Product Description

Description

- Heat Resistance

Applications

- Electric & Electronic Housing (Frame etc),
- Automotives Interior & Exterior Housing (lamp housing etc)

General

Material Status	• Commercial: Active
Literature ¹	• Technical Datasheet - ASTM (English) • Technical Datasheet - ISO (English)
UL Yellow Card ²	• E248280-462768 • E67171-248410
Search for UL Yellow Card	• LG Chem Ltd. • LG ABS
Availability	• Asia Pacific • Europe • Latin America • North America
Features	• Good Heat Resistance
Uses	• Automotive Exterior Parts • Automotive Interior Parts • Electrical Housing
Processing Method	• Injection Molding
Multi-Point Data	• Specific Heat vs. Temperature (ISO 11403-2)

Physical	Nominal Value Unit	Test Method
Density / Specific Gravity -- ⁴	1.05 g/cm ³	ASTM D792
23°C	1.05 g/cm ³	ISO 1183
Melt Mass-Flow Rate (MFR) (220°C/10.0 kg)	9.0 g/10 min	ASTM D1238
Melt Volume-Flow Rate (MVR) (220°C/10.0 kg)	9.0 cm ³ /10min	ISO 1133
Molding Shrinkage - Flow 23°C, 3.20 mm, Injection Molded	0.40 to 0.70 %	ASTM D955
23°C, 3.20 mm ⁵	0.40 to 0.70 %	ISO 294-4

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus 23°C, 3.20 mm, Injection Molded ⁶	2400 MPa	ASTM D638
23°C, 4.00 mm, Injection Molded	2300 MPa	ISO 527-1/50
Tensile Strength Yield, 23°C, 3.20 mm, Injection Molded ⁶	49.0 MPa	ASTM D638
Yield, 23°C, 4.00 mm, Injection Molded	49.0 MPa	ISO 527-2/50
Tensile Strain Yield, 23°C, 4.00 mm, Injection Molded	> 5.0 %	ISO 527-2/50
Break, 23°C, 3.20 mm, Injection Molded ⁶	> 15 %	ASTM D638
Break, 23°C, 4.00 mm, Injection Molded	> 15 %	ISO 527-2/50
Flexural Modulus 23°C, 3.20 mm, Injection Molded ⁷	2550 MPa	ASTM D790
23°C, 4.00 mm, Injection Molded ⁸	2350 MPa	ISO 178
Flexural Strength 23°C, 3.20 mm, Injection Molded ⁷	79.0 MPa	ASTM D790
23°C, 4.00 mm, Injection Molded ⁸	75.0 MPa	ISO 178



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Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength ⁹		ISO 179/1eA
-30°C, Injection Molded	9.0 kJ/m ²	
23°C, Injection Molded	23 kJ/m ²	
Notched Izod Impact		
-30°C, 3.20 mm, Injection Molded	90 J/m	ASTM D256
-30°C, 6.40 mm, Injection Molded	80 J/m	ASTM D256
23°C, 3.20 mm, Injection Molded	270 J/m	ASTM D256
23°C, 6.40 mm, Injection Molded	250 J/m	ASTM D256
-30°C, Injection Molded ⁹	8.0 kJ/m ²	ISO 180/1A
23°C, Injection Molded ⁹	23 kJ/m ²	ISO 180/1A
Hardness	Nominal Value Unit	Test Method
Rockwell Hardness		
R-Scale, 23°C, Injection Molded	110	ASTM D785
R-Scale	112	ISO 2039-2
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load		
1.8 MPa, Unannealed, 6.40 mm, Injection Molded ¹⁰	97.0 °C	ASTM D648
1.8 MPa, Unannealed, 4.00 mm ¹¹	88.0 °C	ISO 75-2/Af
Vicat Softening Temperature	106 °C	ISO 306/B50 ASTM D1525 ¹²
CLTE		ISO 11359-2
Flow : 23 to 60°C	8.0E-5 to 9.0E-5 cm/cm/°C	
Transverse : 23 to 60°C	8.0E-5 to 9.0E-5 cm/cm/°C	
RTI Elec	60.0 °C	UL 746B
RTI Imp	60.0 °C	UL 746B
RTI Str	60.0 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Volume Resistivity (23°C)	1.5E+3 ohms·cm	ASTM D257
Dielectric Strength (23°C, 1.00 mm)	33 kV/mm	ASTM D149
Arc Resistance	PLC 6	ASTM D495
Flammability	Nominal Value Unit	Test Method
Flame Rating		UL 94
1.5 mm	HB	
3.0 mm	HB	
Injection	Nominal Value Unit	
Drying Temperature	80 to 90 °C	
Drying Time	3.0 to 4.0 hr	
Suggested Max Moisture	0.050 %	
Rear Temperature	180 to 210 °C	
Middle Temperature	210 to 230 °C	
Front Temperature	230 to 240 °C	
Nozzle Temperature	230 to 240 °C	
Processing (Melt) Temp	230 to 260 °C	
Mold Temperature	40 to 60 °C	
Back Pressure ¹³	0.981 to 2.94 MPa	



Notes

- ¹ These links provide you with access to supplier literature. We work hard to keep them up to date; however you may find the most current literature from the supplier.
- ² A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.
- ³ Typical properties: these are not to be construed as specifications.
- ⁴ 23°C
- ⁵ Injection molded
- ⁶ 50 mm/min
- ⁷ 15 mm/min
- ⁸ 2.0 mm/min
- ⁹ 4mm
- ¹⁰ Edgewise
- ¹¹ Injection Molded
- ¹² Rate A (50°C/h), Loading 2 (50 N)
- ¹³ Hydraulic Type



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Where to Buy

Supplier

LG Chem Ltd.

Englewood Cliffs, Englewood Cliffs USA

Telephone: 201-816-2302

Web: <http://www.chemwide.com/>

Distributor

CCC Plastics

Telephone: 800-461-1638

Web: <https://www.ccc-group.com/>

Availability: Canada

Channel Prime Alliance

Telephone: 800-247-8038

Web: <http://www.channelpa.com/>

Availability: North America

Chase Plastic Services, Inc.

Chase Plastics Services is a North American distributor with representatives throughout the region. Please find your rep here: <http://www.chaseplastics.com/contact/locations>

Telephone: 800-232-4273

Web: <http://www.chaseplastics.com/>

Availability: North America

Distrupol Ltd

Distrupol Ltd is a Pan European distribution company. Contact Distrupol Ltd for availability of individual products by country.

Telephone: 08452003040

Web: <http://www.distrupol.com/>

Availability: Belgium, Denmark, Finland, Luxembourg, Netherlands, Norway, Sweden

Entec Polymers Latin America

Contact Entec Polymers for availability of individual products by country.

Web: <https://www.entecpolymers.com/>

Availability: Latin America

POLYMIX

POLYMIX is a Pan European distribution company. Contact POLYMIX for availability of individual products by country.

Telephone: +33-3-8920-1380

Web: <http://www.polymix.eu/>

Availability: Belgium, Croatia, France, Luxembourg, Netherlands, Romania, Serbia, Slovenia, Spain, Tunisia

Resin Resource, Inc.

Telephone: 877-652-3431

Web: <http://www.resinresourceinc.com/>

Availability: North America

RESINEX Group

RESINEX is a Pan European distribution company. Contact RESINEX for availability of individual products by country.

Telephone: +32-14-672511

Web: <http://www.resinex.com/>

Availability: Europe

SNETOR Distribution

SNETOR Distribution is a Pan European distribution company. Contact SNETOR for availability of individual products by country.

Telephone: +33-1-4904-8888

Web: <http://www.snetor.com/>

Availability: France

